

Solar cells



Solar Cells

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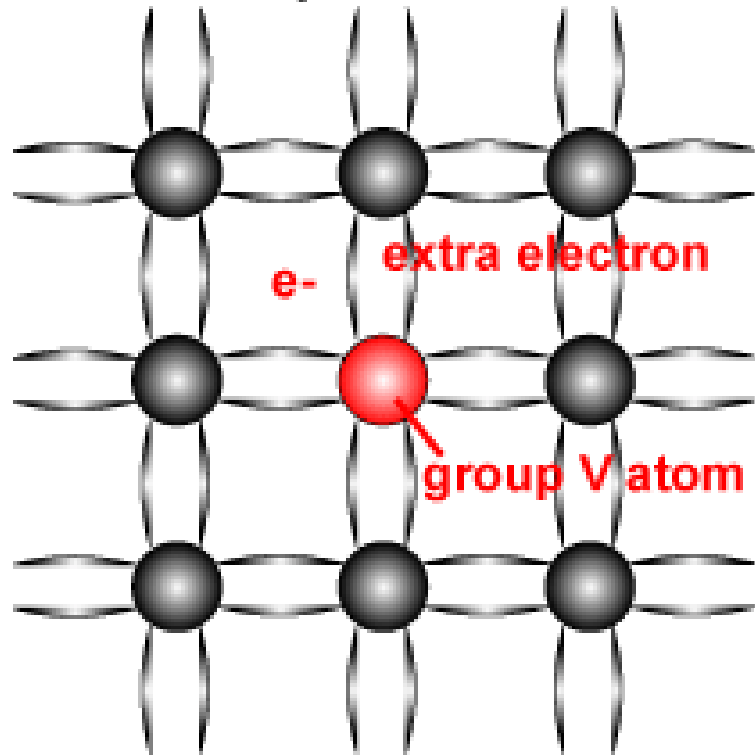
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Pn junction

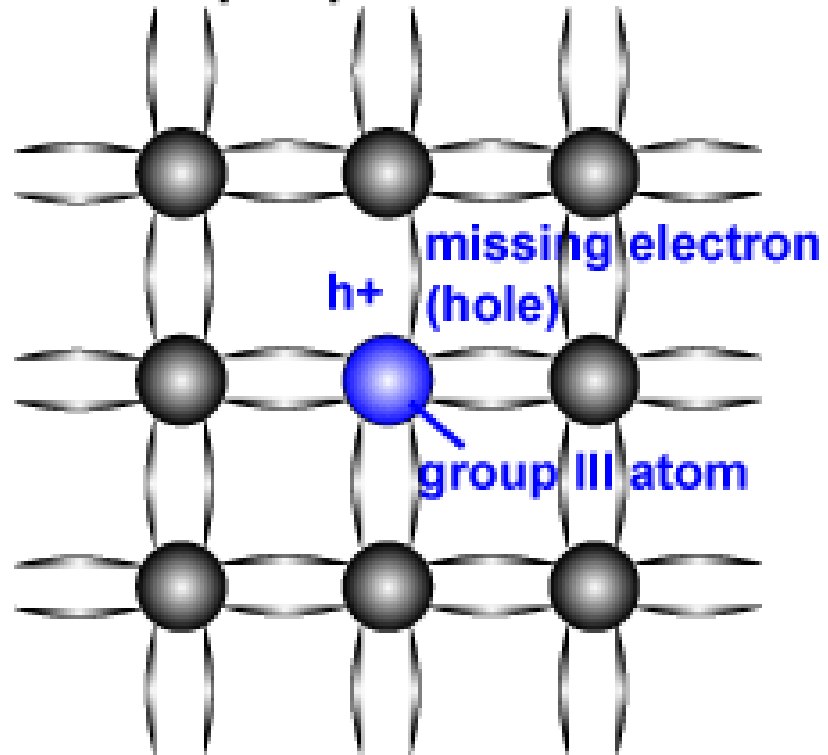
- ▶ Doping is a technique used to vary the number of electrons and holes in semiconductors.
- ▶ Doping creates N-type material when semiconductor materials from Elements pentavalent such as Phosphorus
- ▶ . P-type materials are created when semiconductor materials from Elements of trivalent such as Boron
- ▶ N-type materials increase the conductivity of a semiconductor by increasing the number of available electrons
- ▶ P-type materials increase conductivity by increasing the number of holes present.

	N-type (negative)	P-type (positive)
Dopant	Group V (e.g. Phosphorous)	Group III (e.g. Boron)
Bonds	Excess Electrons	Missing Electrons (Holes)
Majority Carriers	Electrons	Hole
Minority Carriers	Holes	Electrons

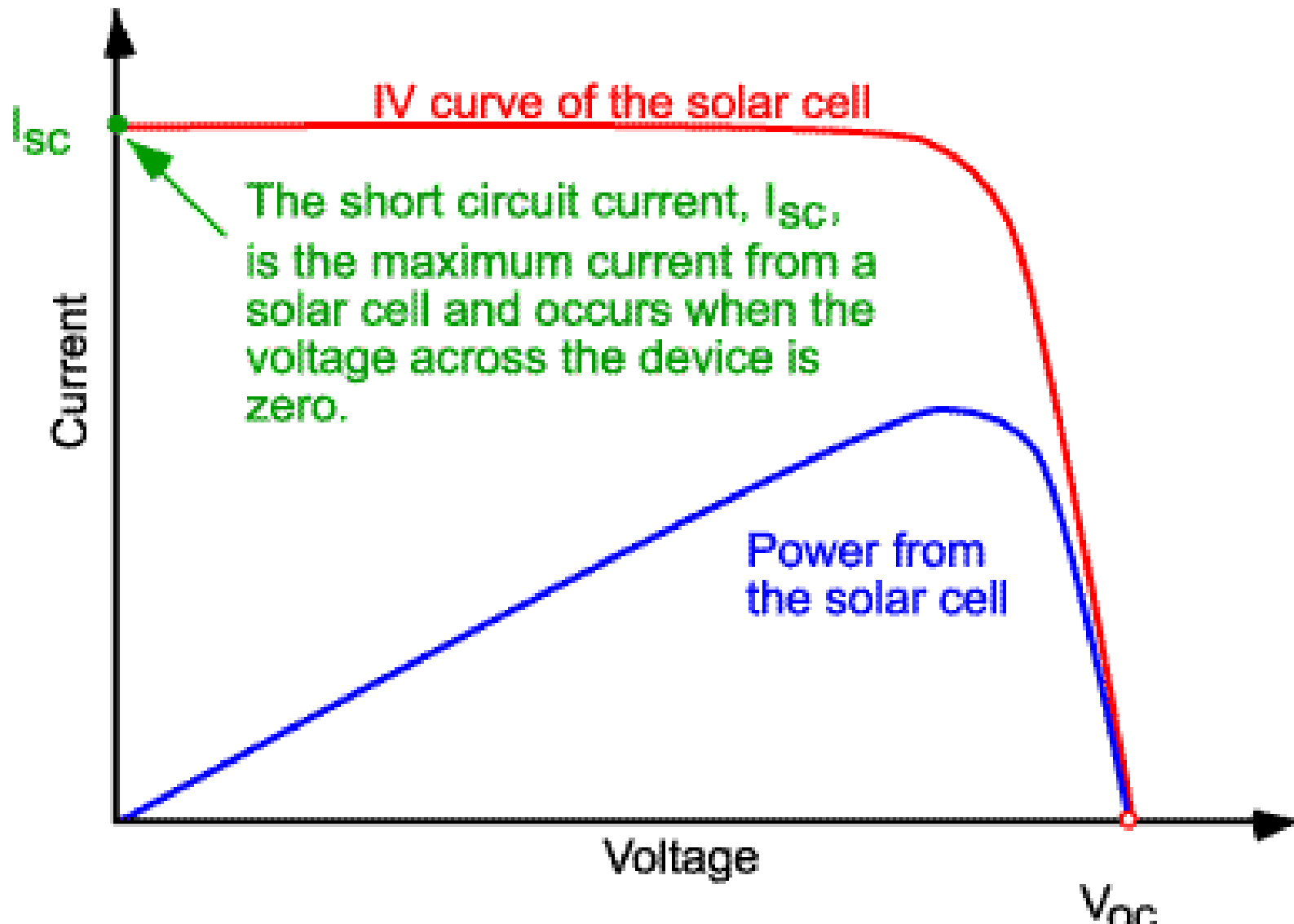
n-doped silicon

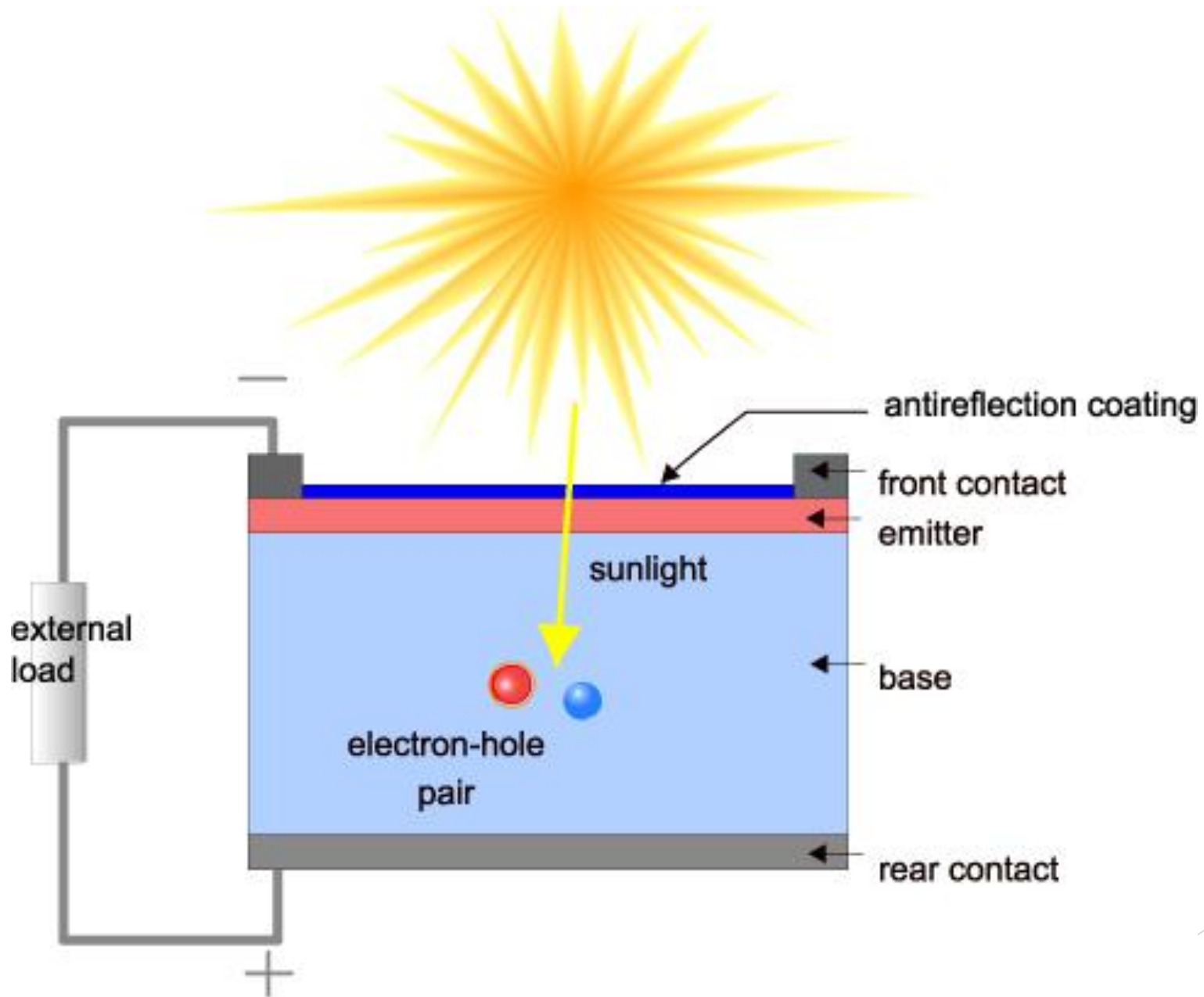


p-doped silicon



IV curve





Anti Reflection Coatings

An antireflection of silicon nitride (Si_3N_4)
or TiO_2

Rear

A thick layer of aluminium paste covers most of the cell and provides a back surface field (BSF). Silver strips are also printed on the cell for later soldering to the interconnect metal tabs.

Crystalline

Monocrystalline

Polycrystalline

**Concentrating
Photovoltaic**

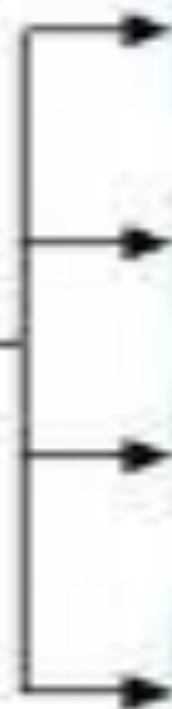
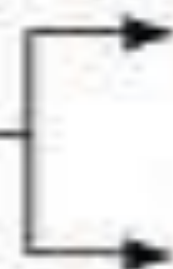
CdTe

CIGS

Thin - film

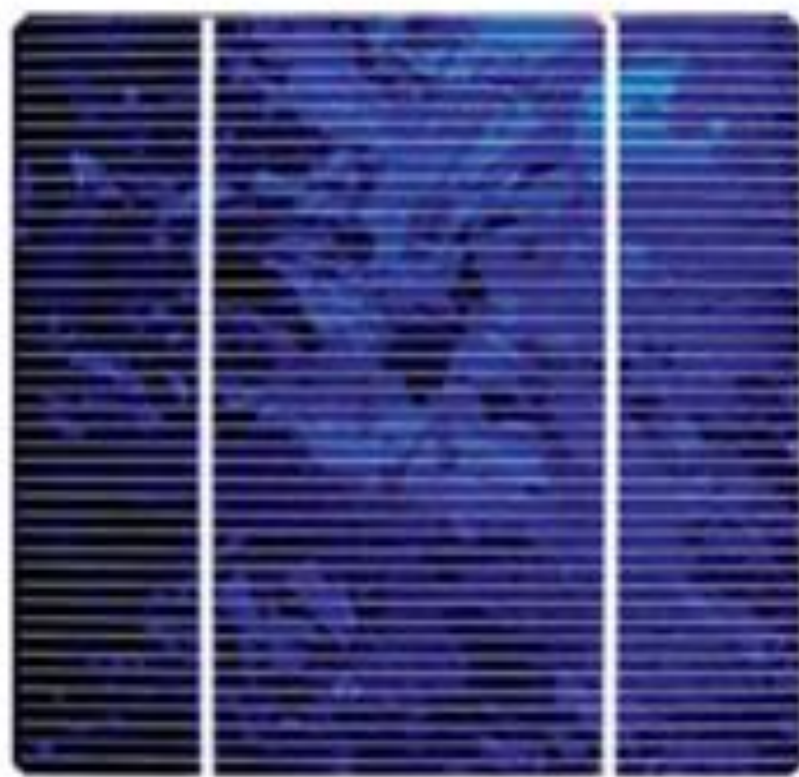
CIS

**Amorphous
Silicon (a-Si)**





Mono



Poly



Thin Film

- **Mono**
Efficiency: from 11-16 %
- **Poly**
Efficiency: from 9-13 %

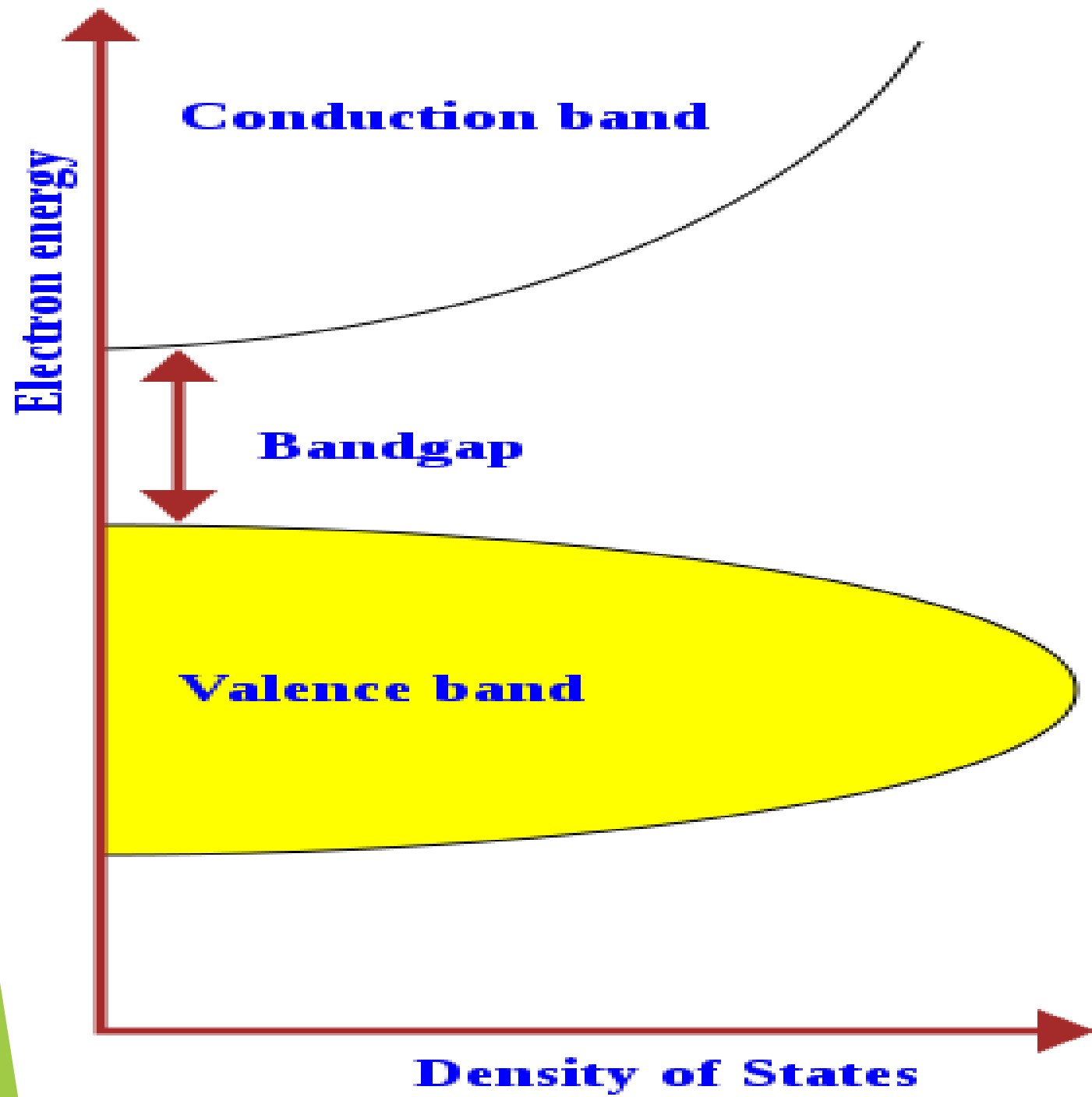
- **A morphs**
Efficiency: from 3-6 %
- **Cadmium Telluride**
(Cd Te)
Efficiency: 19.6 %
in 2013
- **Gallium Arsenide**(Ga
As)

Efficiency: from 25 %

The band gap of a semiconductor is the minimum energy required to excite an electron that is stuck in its bound state into a free state where it can participate in conduction

$$E = h \nu$$

Band Gap



Band Gap

- ▶ Silicon = 1.12 eV
- ▶ CdTe = 1.5 eV
- ▶ GaAs = 1.424 eV